

simultaneous connections with a very low memory footprint of about 2.5MB. This is all possible due to its asynchronous, event driven nature. Nginx supports a large number of Web server languages, but it doesn't have any native support and thus third-party modules have to be used; for instance, it requires PHP-FPM to be installed separately to process PHP scripts. Even though Nginx is used as a standalone Web server, nowadays many developers use it as a static content server in front of the actual server. Some sites dynamically handle the requests at the back-end, and cache them for Nginx to serve when the content is requested by the user.

Nginx can be deployed to serve dynamic HTTP content on the network using FastCGI, SCGI handlers for scripts, WSGI application servers or Phusion Passenger modules, and it can serve as a software load balancer.

There are two versions of Nginx — OSS Nginx and Nginx Plus. The latter offers additional features not included in OSS Nginx, such as active health checks, session persistence based on cookies, DNS-service-discovery integration, a cache purging API, AppDynamic, Datalog, Dynatrace New Relic plugins, Active-Active HA with configuration synchronisation, on-the-fly (with zero downtime) updates for upstream configurations, as well as key value stores using the Nginx Plus API and Web application firewall (WAF) dynamic module.

Nginx was designed to outperform the Apache server by serving static files, using less memory and handling four times as many requests per second.

Features

- Handles 10,000 concurrent connections with low memory; handles static, index files and offers auto-indexing.
- Name- and IP address based virtual servers.
- IPv6 compatible.

- Supports TLS/SSL with SNI, offers OCSP stapling support, gRPC support and HTTP/2 support.
- Configuration upgrade and execution is done without client connection loss. Supports SMTP, POP3 and IMAP proxy.
- Reverse proxy with caching.
- Modular architecture: Filters include gzipping, byte ranges, chunked responses, XSLT, SSI, and an image transformation filter. Multiple SSI inclusions within a single page can be processed in parallel if they are handled by proxied or FastCGI/uwsgi/SCGI servers.

Latest version: 1.17.6

Official website: <https://www.nginx.com/>

H2O Web server

H2O is a new generation HTTP server that has full-featured HTTP/2 implementations of all the current Web servers in use. With H2O as the Web server, users can take advantage of new features of the HTTP/2 specifications, like latency optimisation, server-push and server-side prioritisation, which can take advantage of modern browser features that are seldom talked about. H2O is powered by *libh2o*, which contains all the Web serving power of H2O itself. *libh2o* is also packaged separately for third party use. It is MIT licensed. H2O is written in C and can also be used as a library.

Features

- Supports HTTP 1.0/1.1/2.0, WebSocket and TLS.
- Supports TCP Fast Open, FastCGI, reverse proxy.
- AEAD ciphers, OCSP stapling, mime-type configuration, forward secrecy, chunked encoding.
- Server push, negotiation methods: NPN, ALPN, upgrade, direct.

Latest version: 2.3.0

Official website: <https://h2o.examp1e.net/>

Caddy

Caddy is an open source, HTTP/2 enabled Web server, designed by Matthew Holt and written in the Go language. It is a strong alternative to Apache Web Server. It is easy to configure and use, and is loaded with the latest features like IPv6, Markdown, WebSockets, FastCGO, templates and other out-of-the-box features. Caddy activates HTTPS by default for sites with qualifying domain names (names for which a TLS certificate can be negotiated via the ACME protocol) and redirects HTTP requests to HTTPS. It obtains certificates as needed during startup and keeps them renewed during the lifetime of the server. An alternate configuration method allows Caddy to obtain certificates only as needed during TLS handshakes rather than at startup, a feature dubbed as 'on-demand TLS'. To enable this feature, the user must specify a maximum number of certificates that can be issued this way. It is available for Windows, MAC, Linux, BSD, Solaris and Android.

Features

- Fast HTTP requests, minimal configuration with fast deployment.
- No installation, portable executables, and runs on multiple CPUs/cores.
- Supports IPv6, creates login custom format.
- Serves FastCGI, has reverse proxy, rewrites and redirects, clean URL and Gzip compression, directory browsing, virtual hosts and headers.
- Supports VirtualHost, load balancing with health checks, URL rewriting, basic access authentication, Gzip compression, file browsing, QUIC, etc.

Latest version: 2.0 beta 12

Official website: <https://caddyserver.com/>